

Total No. of Questions :10]

SEAT No. :

P3269

[5670]-538

[Total No. of Pages :5

B.E. (Mechanical)

OPERATION RESEARCH

(2015 Pattern) (Semester-I) (Elective-II) (402045B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Answers in one answer books.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1) a) Discuss various phases of solving Operation research (OR) Problem.[4]
- b) The data refers to the machining whose cost is Rs. 60,000. The other costs are as follows:

| Year                 | 1      | 2      | 3      | 4      | 5      |
|----------------------|--------|--------|--------|--------|--------|
| Resale value (Rs.)   | 42,000 | 30,000 | 20,400 | 14,400 | 9,650  |
| Cost of Spares (Rs.) | 4,000  | 4,270  | 4,880  | 5,700  | 6,800  |
| Cost of labors (Rs.) | 14,000 | 16,000 | 18,000 | 21,000 | 25,000 |

[6]

Determine at which time it is profitable to replace the machine.

OR

- Q2) A department of company has five employees with five jobs to be performed. The time (in hours) that each man takes to perform each job is given in the effectiveness matrix. How should the jobs be allocated, one per employee so as to minimize the total man-hours? [10]

|      |   | Employees |    |     |    |    |
|------|---|-----------|----|-----|----|----|
|      |   | I         | II | III | IV | V  |
| Jobs | A | 10        | 5  | 13  | 15 | 16 |
|      | B | 3         | 9  | 18  | 13 | 6  |
|      | C | 10        | 7  | 2   | 2  | 2  |
|      | D | 7         | 11 | 9   | 7  | 12 |
|      | E | 7         | 9  | 10  | 4  | 12 |

P.T.O.

a) Maxi-min criteria,  
b) Maxi-max criteria,  
c) Mini-max criteria  
d) Laplace criteria

[10]

OR

Distribution centre 3 : 3 million liters, and Distribution centre 4 : 2 million liters, Cost (in hundreds of rupees) of shipping one million litre from each plant to each distribution centre is given in the following table: **[10]**

|        |    | Distribution Centre |    |    |    |
|--------|----|---------------------|----|----|----|
|        |    | D1                  | D2 | D3 | D4 |
| Plants | P1 | 2                   | 3  | 11 | 7  |
|        | P2 | 1                   | 0  | 6  | 1  |
|        | P3 | 5                   | 8  | 15 | 9  |

Find initial feasible solution for a given problem by using

- North-west corner rule
- Least cost method
- Vogel's approximation method

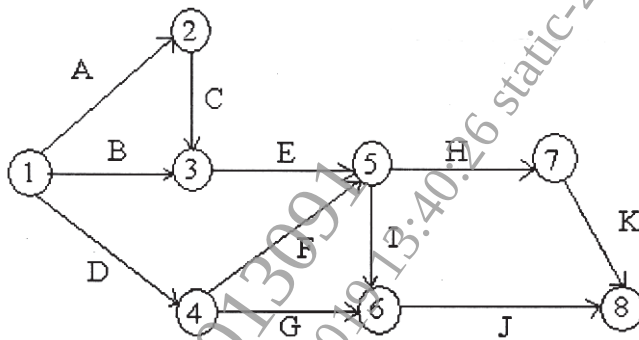
- Q5)** a) Explain with the help of neat sketch looping and dangling type of errors in the network. **[4]**
- b) A company has decided to modernized its office. The major elements of the project are as follows. **[12]**

| Activity | Description                   | predecessor Activity | Duration(Days) |
|----------|-------------------------------|----------------------|----------------|
| A        | Organize sale office          | ---                  | 6              |
| B        | Hire salesman                 | A                    | 4              |
| C        | Train salesman                | B                    | 7              |
| D        | Select advertising agency     | A                    | 2              |
| E        | Plan advertising campaign     | D                    | 4              |
| F        | Conduct advertising campaign  | E                    | 10             |
| G        | Design package                | ---                  | 2              |
| H        | Set up packaging facilities   | G                    | 10             |
| I        | Package initial stocks        | J,H                  | 6              |
| J        | Order stock from manufacturer | ----                 | 13             |
| K        | Select distributors           | A                    | 9              |
| L        | Sell to distributors          | C,K                  | 3              |
| M        | Ship stocks to distributors   | I,L                  | 5              |

- Draw an arrow diagram for this project.
- Find out the critical path
- For each non-critical activity find out the total and free floats.

OR

**Q6) a)** For the project



|                   |   |    |    |   |    |    |    |   |   |    |    |
|-------------------|---|----|----|---|----|----|----|---|---|----|----|
| Task :            | A | B  | C  | D | E  | F  | G  | H | I | J  | K  |
| Least time:       | 4 | 5  | 8  | 2 | 4  | 6  | 8  | 5 | 3 | 5  | 6  |
| Greatest time:    | 8 | 10 | 12 | 7 | 10 | 15 | 16 | 9 | 7 | 11 | 13 |
| Most likely time: | 5 | 7  | 11 | 3 | 7  | 9  | 12 | 6 | 5 | 8  | 9  |

Find the earliest and latest expected time to each event and also critical path in the network. [10]

- b) Compare and contrast CPM and PERT. Under what conditions would you recommend the scheduling by PERT? Justify your answer with reasons. [6]

**Q7) a)** Six jobs are to be process on three machines. The processing time is as follows, Find the optimal schedule so that the total elapsed time is minimized. [10]

|                  |   |   |   |    |   |   |    |
|------------------|---|---|---|----|---|---|----|
| Job              | A | B | C | D  | E | F | G  |
| Machine M1 (Min) | 3 | 8 | 7 | 4  | 9 | 8 | 7  |
| Machine M2 (Min) | 4 | 3 | 2 | 5  | 1 | 4 | 3  |
| Machine M3 (Min) | 6 | 7 | 5 | 11 | 5 | 6 | 12 |

- b) Explain with the help of neat sketch a generalised queuing model. Also explain the Kendall's notation for representing queuing models? [6]

OR

**Q8) a)** In a railway marshalling yard, goods trains arrive at a rate of 30 trains per day. Assuming the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 36 minutes. Calculate:

- Expected queue size (Line length)
- Probability that the queue size exceeds if the input of trains increases to an average of 33 per day. What will be the change in 1 and 2. [8]

- b) Seven jobs are performed, first on machine X and then on machine Y. The time taken, in hours by each job on each machine is given below: Determine a sequence and the idle time on both the machine. [8]

| Job       | J1 | J2 | J3 | J4 | J5  | J6 | J7 |
|-----------|----|----|----|----|-----|----|----|
| MachineX  | 20 | 90 | 80 | 20 | 120 | 15 | 65 |
| Machine Y | 25 | 60 | 75 | 30 | 90  | 35 | 50 |

- Q9) a)** Solve the following integer LP problem using the cutting plane method  
Maximize  $Z=3X_1+12X_2$  [12]

Subject to constrain

i)  $2X_1+4X_2 \leq 7$

ii)  $5X_1+3X_2 \leq 15$

iii)  $X_1, X_2 \geq 0$  and are integers.

- b) Explain in brief Branch and Bound method. [6]

OR

- Q10)a)** A company has five salesmen who have to be allocated to four marketing zones. The return (Profit) from each zone depends upon the numbers of salesman working in that zone. The expected return for different numbers of salesman in different zones, as estimated from the past record, are given in the following table. Determine the optimum allocation policy. (Use DP) [12]

| Number of salesman | Marketing Zones |        |        |
|--------------------|-----------------|--------|--------|
|                    | Zone 1          | Zone 2 | Zone 3 |
| 0                  | 30              | 35     | 42     |
| 1                  | 45              | 45     | 54     |
| 2                  | 60              | 52     | 60     |
| 3                  | 70              | 64     | 70     |
| 4                  | 79              | 72     | 82     |
| 5                  | 90              | 82     | 95     |
| 6                  | 98              | 93     | 102    |
| 7                  | 105             | 98     | 110    |
| 8                  | 100             | 100    | 110    |
| 9                  | 90              | 100    | 110    |

- b) Explain in brief Dynamic programming (DP) model. [6]

